

- Q1. A cap has an original price of 15 dollars. There is a 5% discount. What is the final amount paid?
- Q2. A cap has an original price of c dollars. There is a 5% discount. What is the final amount paid?

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Q5. A cap has an original price of c dollars. There is a 5% discount. The final amount paid is 15 dollars. What is the original price?

Question 5 says a cap has an original price of c dollars. There is a 5 percent discount. The final amount paid is 15 dollars. What is the original price? We have the same information as in question 2, and the new info that the final amount paid is 15 dollars. Our answer to question 2 was c minus 0.05 times c , which is **one** way to write the final amount paid. This question tells us the final amount paid can **also** be 15, so we set these equal. In other words, since c minus 0.05 times c is the final amount paid, and since 15 is **also** the final amount paid, we set these equal, since both represent final amount paid. We can factor out c from the left side, which gives us 1 minus 0.05 times c equals 15 . We divide both sides by 1 minus 0.05 . So we get c is equal to 15 divided by 1 minus 0.05 . This is not simplified, but this unsimplified answer is a number that answers question 5. Question 6 says a cap has an original price of c dollars. There is a 5 percent tax. The final amount paid is 15 dollars. What is the original price? The answer to question 4 was c plus 0.05 c , which represented the final amount paid. Question 6 tells us that 15 **also** represents the final amount paid. So we set these two expressions equal to create the equation c plus 0.05 c equals 15 . Factor c , so 1 plus 0.05 times c equals 15 . Divide both sides by 1 plus 0.05 , so c equals 15 divided by 1 plus 0.05 .

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Q7. A meal has a price of m dollars before tax and tip. There is a 5% tax. On the post-tax amount, the customer leaves a 15% tip. The final amount paid is 38 dollars. What was the pre-tax pre-tip price of the meal?

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